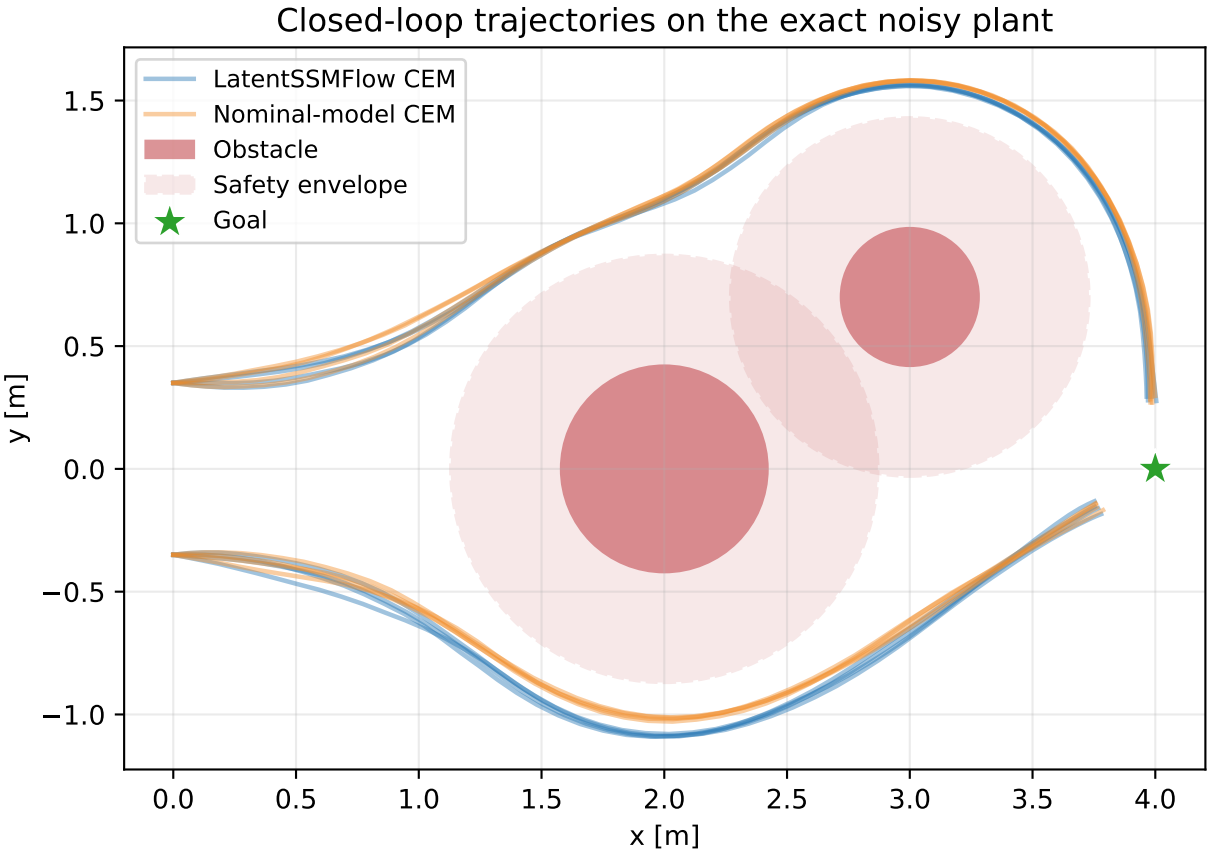
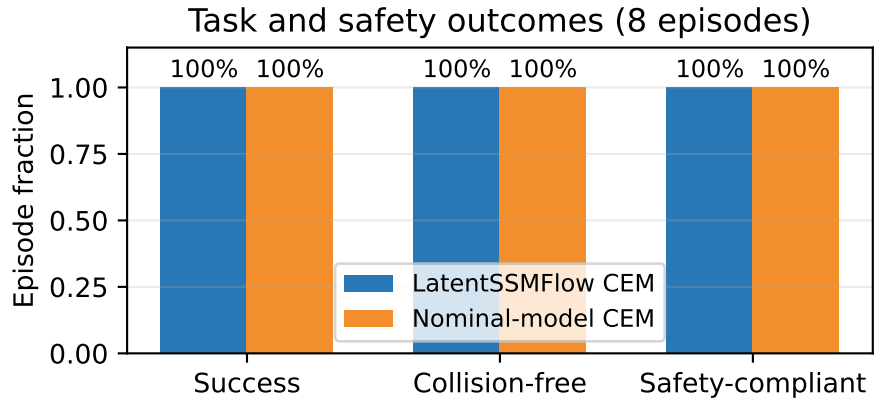
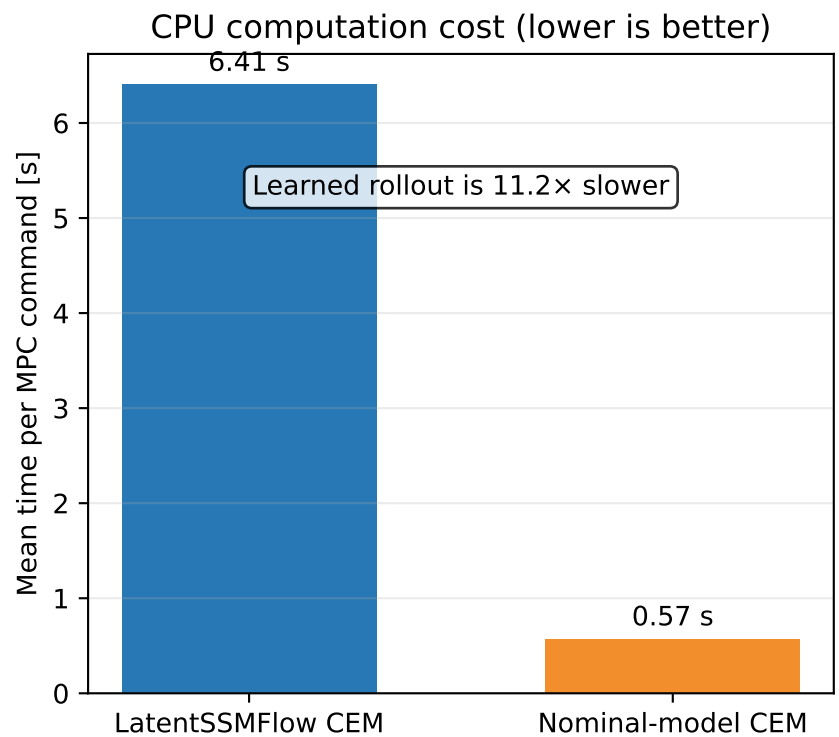

Closed-loop trajectories on the exact noisy plant
LatentSSMFlow CEM
Nominal-model CEM
Obstacle
Safety envelope
Goal
y [m]
x [m]
1.5
1.0
0.5
0.0
−0.5
−1.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0


Task and safety outcomes (8 episodes)
Episode fraction
1.00
0.75
0.50
0.25
0.00
100%
100%
100%
100%
100%
100%
LatentSSMFlow CEM
Nominal-model CEM
Success
Collision-free
Safety-compliant


CPU computation cost (lower is better)
Mean time per MPC command [s]
6.41 s
Learned rollout is 11.2× slower
0.57 s
0
1
2
3
4
5
6
LatentSSMFlow CEM
Nominal-model CEM